

## Calibration results

=====

### Normalized Residuals

-----  
Reprojection error (cam0): mean 0.319224845056, median 0.290443904552, std: 0.184609763045  
Reprojection error (cam1): mean 0.313275900772, median 0.288779200388, std: 0.171895435968  
Gyroscope error (imu0): mean 0.632702552371, median 0.555015341365, std: 0.37480123449  
Accelerometer error (imu0): mean 0.408909229782, median 0.372970019943, std: 0.21819321908

### Residuals

-----  
Reprojection error (cam0) [px]: mean 0.319224845056, median 0.290443904552, std: 0.184609763045  
Reprojection error (cam1) [px]: mean 0.313275900772, median 0.288779200388, std: 0.171895435968  
Gyroscope error (imu0) [rad/s]: mean 0.00912496597876, median 0.00800454508784, std: 0.0054054602  
Accelerometer error (imu0) [m/s^2]: mean 0.0589737277996, median 0.0537905012446, std: 0.031468273

### Transformation (cam0):

-----  
T\_ci: (imu0 to cam0):

```
[[ 0.99999049 -0.00112537 0.00421247 -0.03701856]
 [-0.0011572 -0.99997075 0.00756041 -0.00375399]
 [ 0.00420384 -0.00756522 -0.99996255 -0.01724135]
 [ 0.         0.         0.         1.        ]]
```

T\_ic: (cam0 to imu0):

```
[[ 0.99999049 -0.0011572 0.00420384 0.03708634]
 [-0.00112537 -0.99997075 -0.00756522 -0.00392597]
 [ 0.00421247 0.00756041 -0.99996255 -0.01705638]
 [ 0.         0.         0.         1.        ]]
```

timeshift cam0 to imu0: [s] ( $t_{imu} = t_{cam} + \text{shift}$ )  
-0.00541137170967

### Transformation (cam1):

-----  
T\_ci: (imu0 to cam1):

```
[ 0.      0.      0.      1.    ]]
```

T\_ic: (cam1 to imu0):

```
[[ 0.99998392 -0.00566713 0.00021646 -0.0026764 ]  
 [-0.0056653 -0.99995434 -0.0076957 -0.00361071]  
 [ 0.00026006 0.00769435 -0.99997036 -0.01687228]  
 [ 0.      0.      0.      1.    ]]
```

timeshift cam1 to imu0: [s] ( $t_{imu} = t_{cam} + \text{shift}$ )

-0.00543720269336

Baselines:

-----

Baseline (cam0 to cam1):

```
[[ 0.99998188 0.00450992 0.00398658 0.03976384]  
 [-0.00450935 0.99998982 -0.00015301 0.00008849]  
 [-0.00398723 0.00013503 0.99999204 0.00019512]  
 [ 0.      0.      0.      1.    ]]
```

baseline norm: 0.0397644205316 [m]

Gravity vector in target coords: [m/s<sup>2</sup>]

[ 0.27448991 -0.09967867 -9.80220089]

Calibration configuration

=====

cam0

-----

Camera model: pinhole

Focal length: [129.3245711597599, 129.0866673590973]

Principal point: [327.7427751834073, 239.1195478617305]

Distortion model: equidistant

Distortion coefficients: [0.47658318447819037, 0.25945944427527967, -0.15717299215212371, 0.037438

Type: aprilgrid

Tags:

Spacing 0.010372 [m]

cam1

-----

Camera model: pinhole

Focal length: [128.73859719304275, 128.52670263092364]

Principal point: [318.62953276393586, 242.69156700829126]

Distortion model: equidistant

Distortion coefficients: [0.5278034980799278, 0.08624865376811669, 0.08769574199799478, -0.0824019

Type: aprilgrid

Tags:

Rows: 7

Cols: 5

Size: 0.02593 [m]

Spacing 0.010372 [m]

IMU configuration

=====

IMU0:

-----

Model: calibrated

Update rate: 208.0

Accelerometer:

Noise density: 0.01

Noise density (discrete): 0.144222051019

Random walk: 0.001

Gyroscope:

Noise density: 0.001

Noise density (discrete): 0.0144222051019

Random walk: 0.0001

T\_i\_b

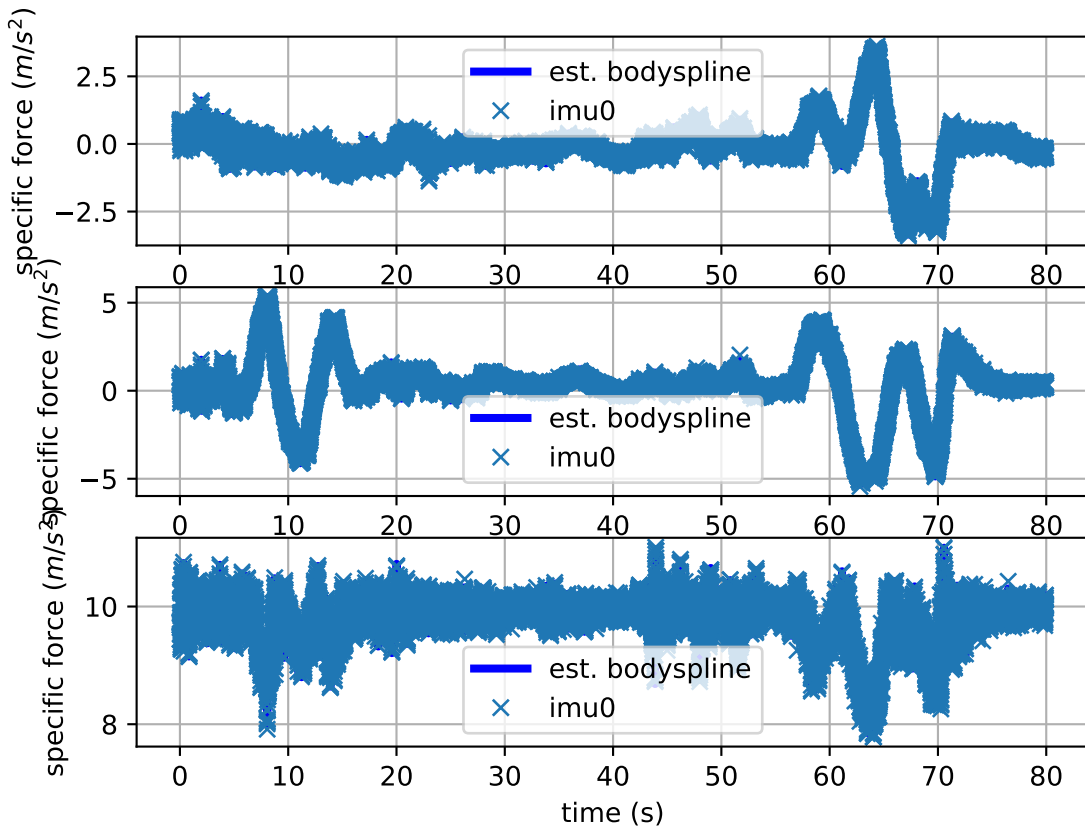
[[ 1. 0. 0. 0.]

[ 0. 1. 0. 0.]

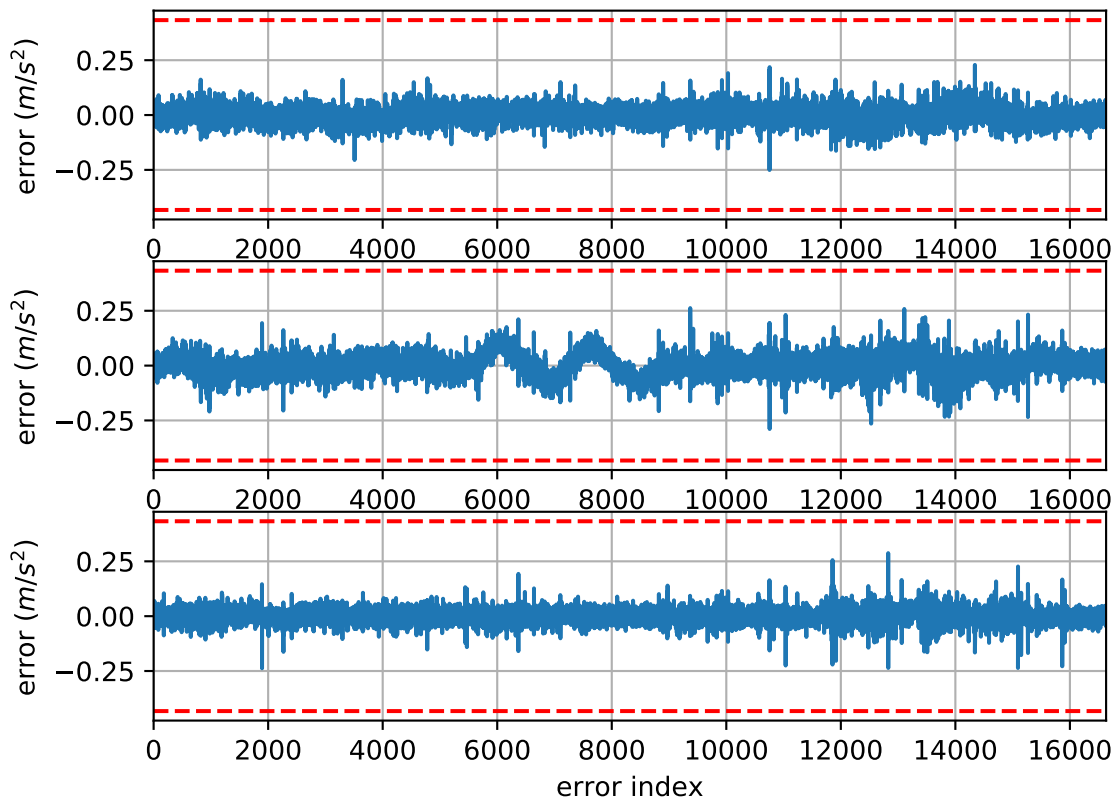




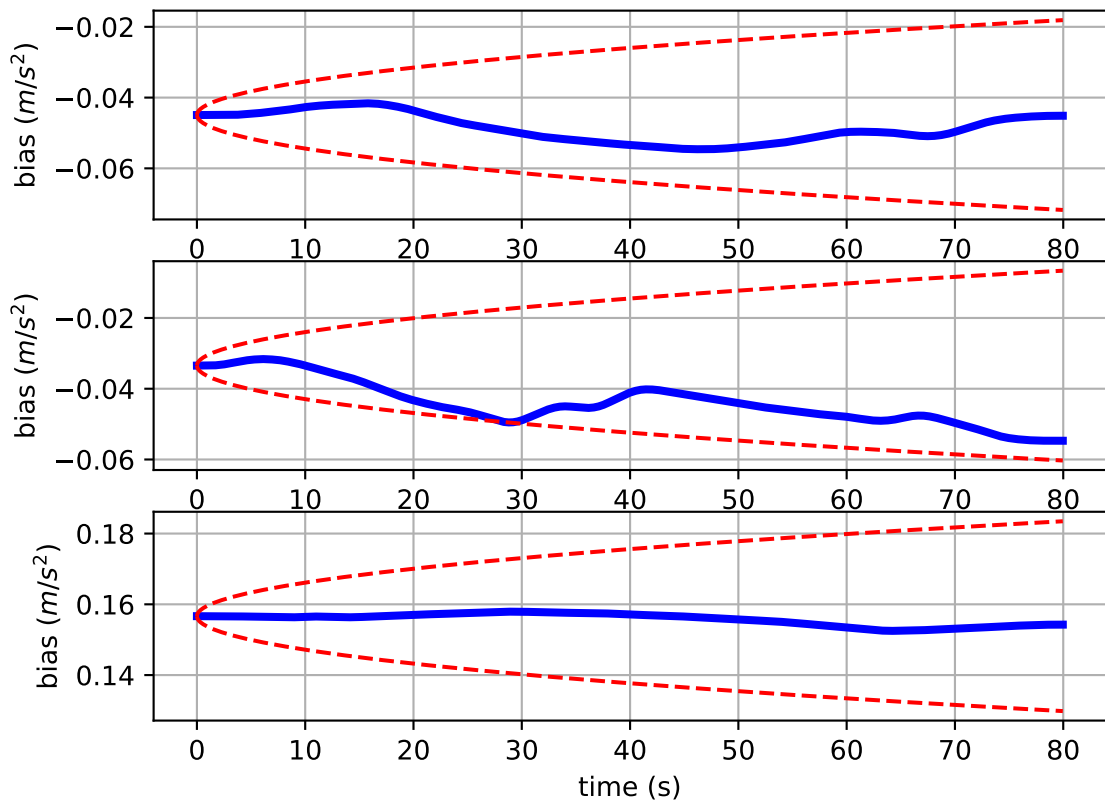
Comparison of predicted and measured specific force (imu0 frame)



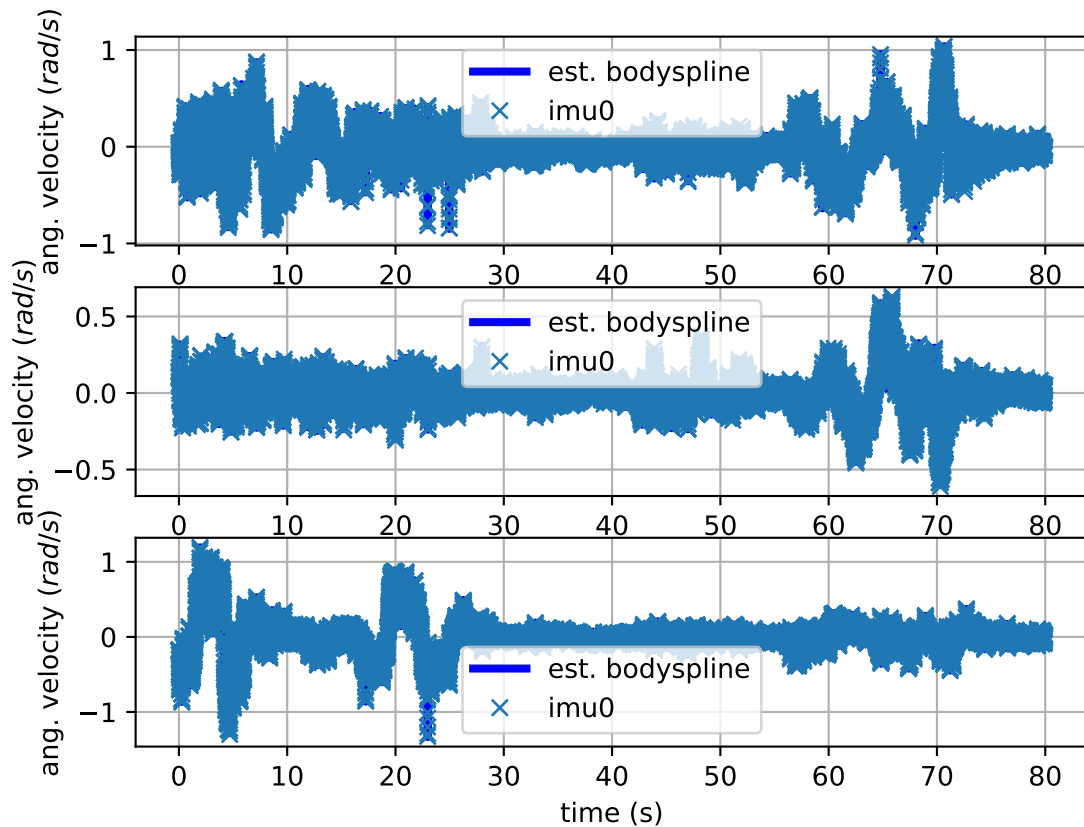
imu0: acceleration error



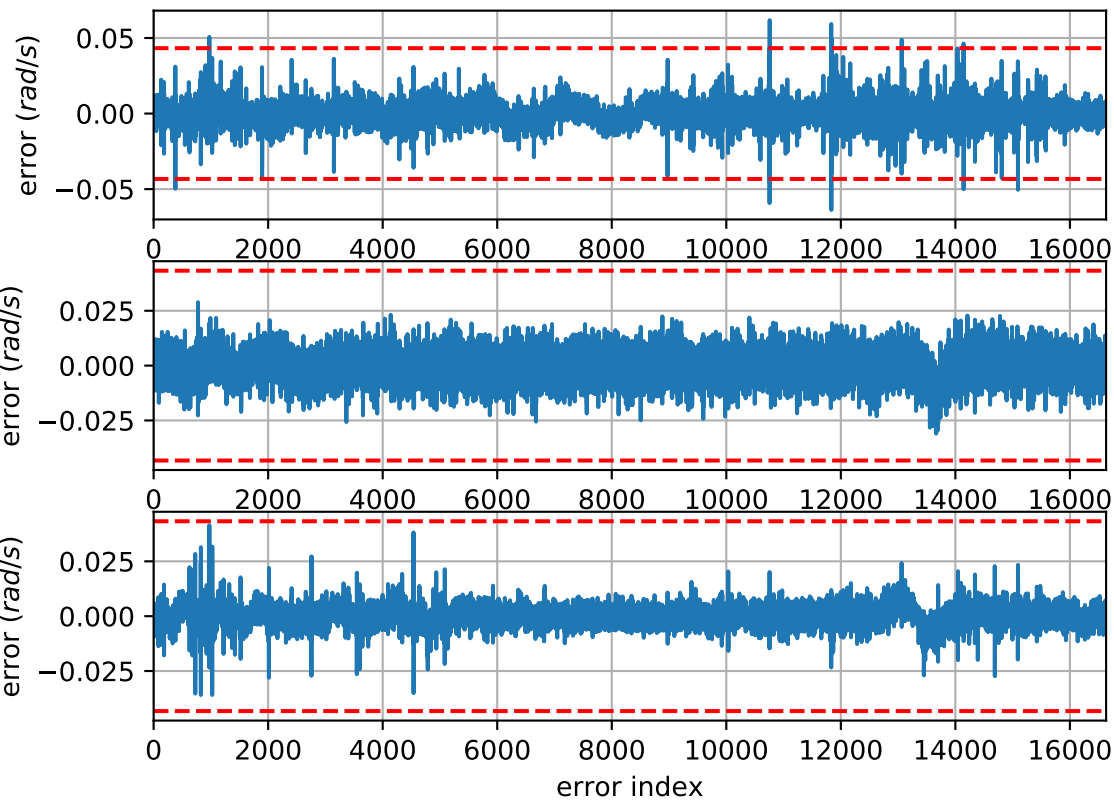
imu0: estimated accelerometer bias (imu frame)



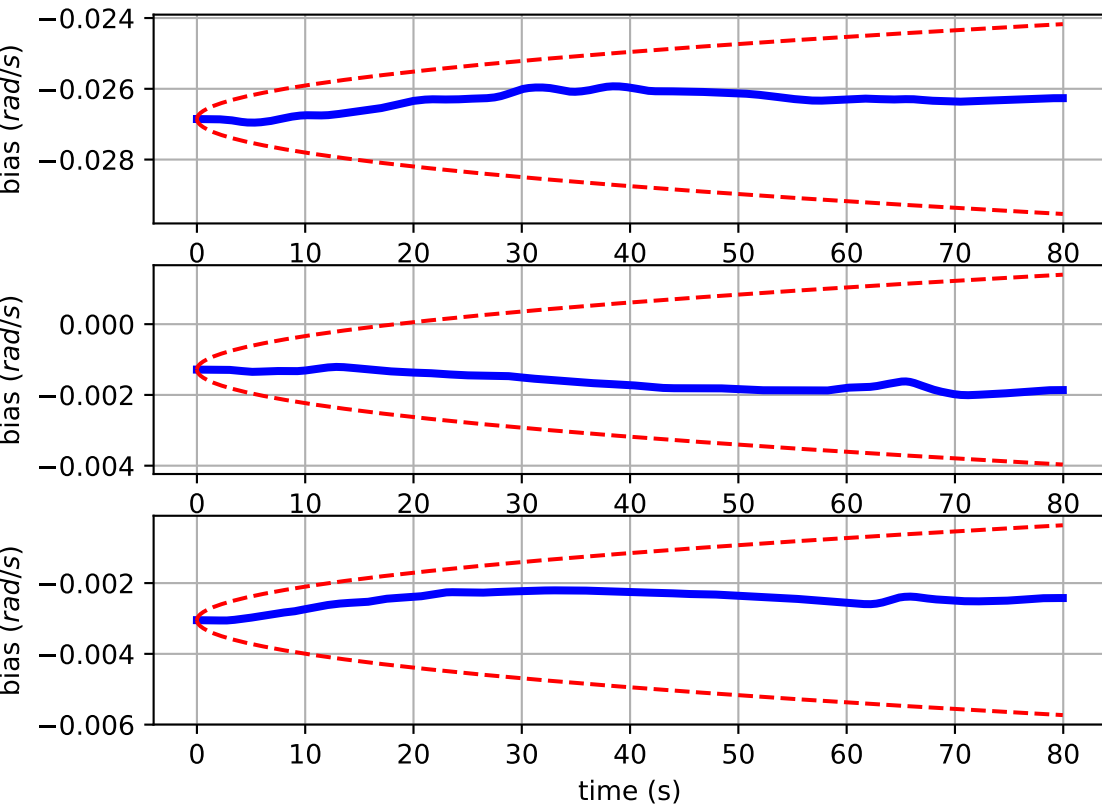
# Comparison of predicted and measured angular velocities (body frame)



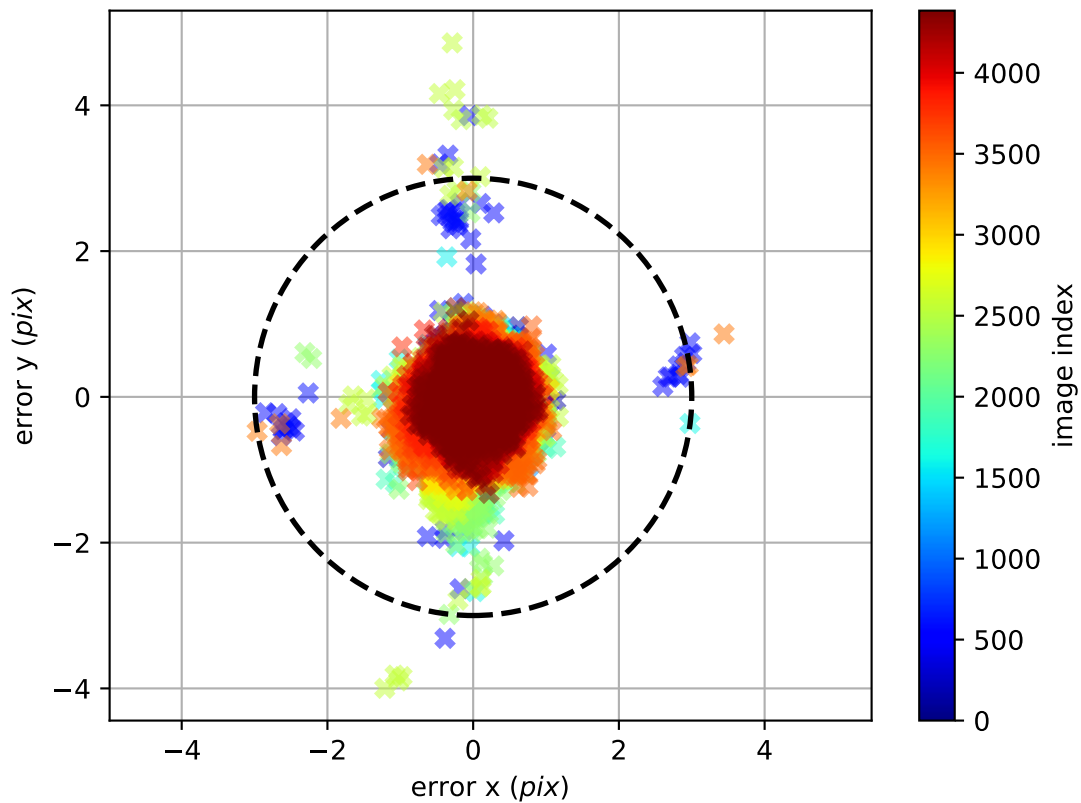
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

